

Turbo Evade: Real Cars Edition – Project Report

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1. Introduction

Turbo Evade: Real Cars Edition is an action-packed HTML5 browser game developed using HTML5, CSS3, and Vanilla JavaScript. The game challenges players to drive through busy highways, avoid traffic, collect rewards, manage fuel, and progress through 150 levels.

2. Objectives

The project aims to create an engaging browser-based racing game without using external game engines. The objectives include responsive gameplay, progressive difficulty, customization options, power-ups, and persistent player progression.

3. Technologies Used

HTML5 for structure, CSS3 for animations and themes, Vanilla JavaScript (ES6+) for game logic, SVG graphics for scalable visuals, and the Web Audio API for sound generation.

4. Key Features

Five customizable car skins, five road themes, 150 unlockable levels, a fuel management system, shield and score multiplier power-ups, dual-currency economy, and a reanimation feature.

5. Gameplay Mechanics

Players move across three lanes using keyboard or touch controls. The score increases continuously during gameplay. Fuel decreases over time and must be replenished. Difficulty increases as players advance through levels.

6. Power-Ups and Collectibles

Gold Balls provide temporary shields, Blue Balls activate a 2× score multiplier, Red Balls cause instant crashes, Green Coins provide session currency, Pink Coins provide persistent currency, and fuel tanks restore fuel.

7. User Interface Design

The game includes a main menu, garage, settings menu, level map, HUD with score and fuel indicators, and a game-over screen with restart and reanimation options.

8. Level Progression System

The game contains 150 levels. The first five levels are unlocked by default. Additional levels can be unlocked using Pink Coins. Vehicle speed and traffic density increase as players progress.

9. Technical Architecture

The game loop uses `requestAnimationFrame` for smooth rendering. Objects are rendered using DOM elements instead of canvas. Collision detection is based on bounding-box overlap checks. Data such as high scores and unlocked levels are stored using `localStorage`.

10. Project Statistics

150 track levels, 5 car skins, 5 road themes, 7 power-up types, 2 currencies, and a JavaScript codebase exceeding 800 lines.

11. Conclusion

Turbo Evade demonstrates how modern browser technologies can be used to develop a complete racing game without external frameworks. The project combines game mechanics, UI design, progression systems, and performance optimization to deliver an engaging gaming experience.